

# TFPT — Frontier items

$\eta_B$ ,  $m_p/m_e$ , the Koide relation, dark matter and quantum gravity —  
the honest status, not forced onto the ladder

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## What this document is about — the status authority for the frontier

The honest frontier: which physics has a genuine TFPT handle and which does not. For each of  $\eta_B$ ,  $m_p/m_e$ , the Koide relation, dark matter and full quantum gravity, this note states the genuine structural handle, the precision with which it currently lands, and — crucially — what is *not* a clean compiler power and is deliberately *not* forced onto the ladder. **This document is the status authority for the frontier items:** where any other document's wording and the markers here disagree, the typing here (and the `status_ledger.csv`) wins.

## The TFPT document set — what is treated where

**Plain language:** TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant  $c_3 = \frac{1}{8\pi}$  and the carrier rank  $g_{\text{car}} = 5$  — build  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and five companions**, best read in order:

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). (*entry point*)
1. **tfpt\_1\_architecture\_e8** — the two axioms, the derivation map, the EM fixed point  $\alpha^{-1}$ , the  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  construction, the QBL theorem chain.
2. **tfpt\_2\_standard\_model** — the SM in one  $\varphi_0$ -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor,  $Q_+$  cohomology), the worked closures.
3. **tfpt\_3\_e8\_audit\_bootstrap** — the seven  $E_8$  slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt\_4\_frontier** — honest status of  $\eta_B$ ,  $m_p/m_e$ , Koide, dark matter, full QG.
5. **tfpt\_5\_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

**Companions:** **R.** **tfpt\_research\_contracts** — the open interfaces ( $v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$ ) as numbered contracts; **H.** **tfpt\_horizon\_readouts** — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** **origin\_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** **tfpt\_safeguards** — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team); **P.** **tfpt\_prime\_front** — research documentation of the prime / zeta line: the seven load-bearing modules on the  $E_8$  glue census (Hecke from geometry through the transport ledger), the sandbox localization program that follows them, and the kill list. Documentation, not a result: it makes no claim about the Riemann Hypothesis.

You are reading document 4 (Frontier).

**TFPT status at a glance — read this card the same way in every document**

**Two inputs, one compiler.** From the boundary constant  $c_3 = \frac{1}{8\pi}$  (axiom P1) and the five-slot carrier  $g_{\text{car}} = 5$  (axiom P2) — jointly the elementary symmetric data of one anchor  $a = (1, 1, 2)$  — the discrete compiler  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ( $\alpha^{-1} = 137.0359992$ ) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

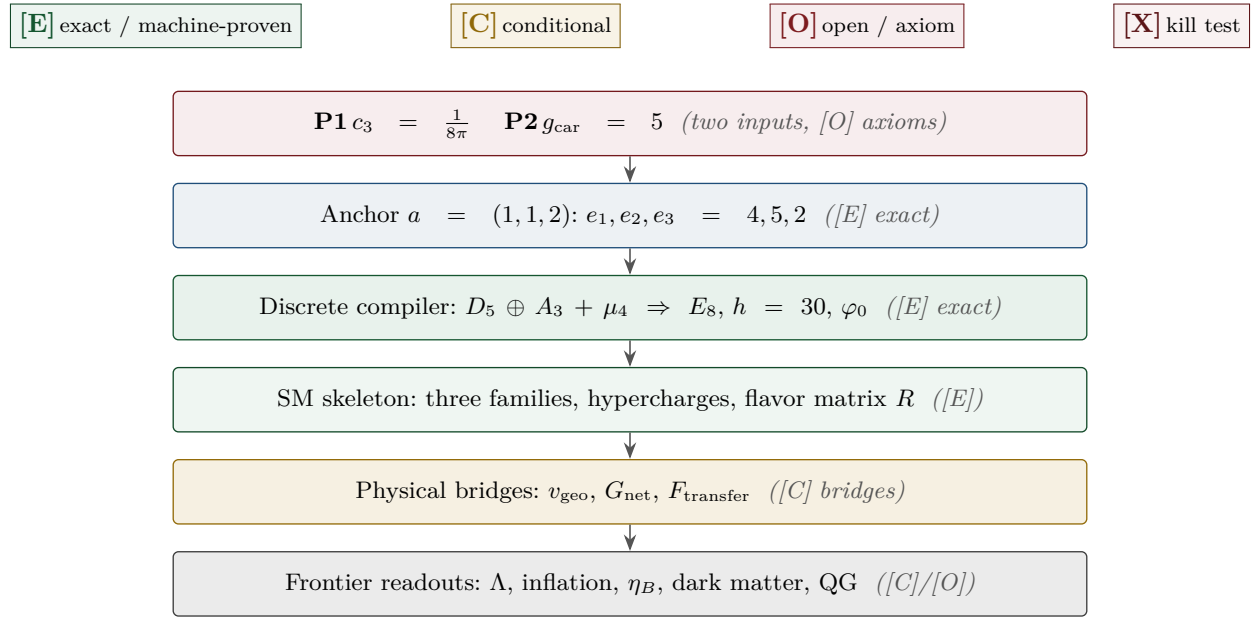
**What is closed, and what genuinely remains.** The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

| Interface             | Question                                                                                        | Status        |
|-----------------------|-------------------------------------------------------------------------------------------------|---------------|
| $v_{\text{geo}}$      | the one metrology unit ( $= 1/\sqrt{G} = m/\mu$ ); No-Unit Thm: no compiler scale               | primitive [O] |
| $G_{\text{net}}$      | simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$ | [C]           |
| $F_{\text{transfer}}$ | one functor, four typed interfaces (Koide, $\eta_B$ , axion, $m_p/m_e$ )                        | [C]           |

**Gravity is parameter-free.** The classical metric-sector field equation is no longer only an  $R+R^2$  readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  with *both* coefficients fixed —  $c_3^{-1} = 8\pi$  (no free dimensionless Newton dial; the thermodynamic origin  $2\pi/\eta$  *coincides* with the geometric one  $|\mathbb{Z}_2|2\pi\chi$  via  $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$ , so  $c_3$  is triply over-determined — anchor, geometry, thermodynamics, v358) and  $\Lambda$  from  $\alpha$  ( $\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$ , v60); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (v359). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in v473 without changing the typing) and the one unit  $v_{\text{geo}}$ ; the global measure (C7 = QG.AMB.01) is now a [C] redundancy (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01 and Bisognano–Wichmann.

The seam-net structural theorem  $G_{\text{net}}$  (= SEAM.EQUIV.01: the raw seam *is* the holomorphic  $(E_8)_1$  net) now carries two named routes (route split 2026-07-22, no status change in substance): the MMST route SEAM.EQUIV.MMST.01 is *closed modulo cited theorems*; the twistor route SEAM.EQUIV.TWISTOR.01 (Costello–Li, prepared by CELEST.SEAM.01) and the unconditional parent stay [O]. On the MMST route: the explicit lattice model (v367/v368) and the  $S_3$  closure stack pin the target at every computable level — central charge  $c = 8$  (v376), the  $(E_8)_1$  character with 248 currents and one primary (v377), genus-one torus GSD = 1 (v378) and reflection positivity (v379) — and it is Lean-formalised as FORM.SEAM.MMST.01: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the **#print axioms** check is clean. The *only* residual of that route that stays [O] is the abstract continuum existence of the scaling limit (v336) — its 128-spinor extension leg since certified at net level by the peer-reviewed crossed-product package ( $h_s=16/16=1 \in \mathbb{Z}$ , the Longo–Rehren locality criterion; Böckenhauer; KLM  $\mu=4/2^2=1$ ; AGT/AMT the independent second witness) with the realisation input reduced to invariant level and the parallel Lean route **seamResidualClosed'** (v469); so the MMST route is *closed modulo cited theorems*, not solved. QGEO.SYM.01 (the  $\mu_4$  deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (v335, Lean FORM.QGEO.BW.01). The nonperturbative quantum-gravity measure QG.AMB.01 is no longer carried as an open item: it is a [C] *redundancy* (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01 and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin  $1.648 > 0$ , v76/v330). So the honest residual is the one unit  $v_{\text{geo}}$  plus the typed  $F_{\text{transfer}}$  interfaces, above the cited-theorem ceiling on the seam and with QG.AMB.01 a [C] redundancy. Everything carries a claim ID resolving to a row of **verification/status\_ledger.csv** (the single source of truth); **if the text and the ledger ever disagree, the ledger wins**. The historical labels ( $U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$ ) are retained only for ledger continuity — the live residual is  $v_{\text{geo}} \oplus F_{\text{transfer}}$ , with the seam  $G_{\text{net}}$  closed modulo cited theorems on its MMST route (parent [O]) and QG.AMB.01 a [C] redundancy.

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in **verification/status\_ledger.csv**.



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**Ground rule.** These five are exactly the items previously flagged “*not on a clean action/ladder yet — do not force.*” This note gives, for each, the *genuine* structural handle that does exist, the precision with which it currently lands, and a clear statement of what is *not* a single compiler power. No fabricated ladder rungs; the honest markers **[E]**/**[C]**/**[O]** are applied throughout. (All numbers machine-verified.)

### Hard rule on the frontier — one interface $F_{\text{transfer}}$

**Koide,  $\eta_B$ , the axion relic scale and  $m_p/m_e$  are *not* compiler powers unless their missing QFT-transfer / cosmology computation is explicitly supplied.** They are deliberately typed interfaces, not structural holes, and they are *four instances of one missing functor*  $F_{\text{transfer}}$  — the continuous transport from compiler *source* data to measured observables (Koide source→pole;  $\eta_B$  source→Boltzmann relic; axion scale→cosmological abundance;  $m_p/m_e \rightarrow$  QCD/EW matching). Each has a genuine handle (a near-value, a determined readout, a fixed candidate) but its *exact* status is conditional on that named transfer calculation — now a runnable **[C]** solver with a kill test (v371–v375), still a transfer target, never a compiler power. In particular  $\eta_B$  is a *transfer target*, not a primitive compiler prediction. They must never be quoted as primitive TFPT outputs.

*Why this is principled, not a gap:* these are precisely the **discrete**↔ **continuous boundary** of the theory — the points where the discrete compiler hands its skeleton off to continuous physics (QCD confinement, cosmological Boltzmann transport, relic integrals). The cleanness of a quantity tracks how far it sits from that handoff: pure compiler ratios are exact, QCD/cosmology-dressed quantities are typed interfaces. [[verification/v35\\_quark\\_qcd\\_boundary.py](#)]

*Machine-enforced.* This rule is a CI-style guard ([[verification/v187\\_ftransfer\\_laws.py](#)]): the four frontier families ( $F_{\text{pole}}$  Koide,  $F_{\text{Boltzmann}}$   $\eta_B$ ,  $F_{\text{relic}}$  axion,  $F_{\text{QCD}}$   $m_p/m_e$ ) are read from the ledger and required to carry a **[C]**/**[O]** marker — exact algebraic sub-parts (the Koide 53/54 factor,  $b_3 = -7$ ) may be **[E]**, but the *physical* prediction never is. No future edit can quietly promote a frontier number to a closed compiler output. A companion *prose* sentinel ([[verification/v188\\_frontier\\_wording\\_guard.py](#)]) forbids stale sentences whose semantics an earlier round superseded, so a frozen release can never re-introduce wording that was true yesterday and is half-true today.

*The functor contract **CONTRACT.F.01**.* Beyond the typing guard, the functor is pinned by four checkable structural axioms ([[verification/v213\\_ftransfer\\_functor.py](#)]): each instance is a *discrete* compiler kernel **[E]** composed with an *external* continuous solver **[C]**, and the functor is (1)  $\mu_4$ -deck equivariant (the Koide multiplier  $\lambda_2 = (2/3)^6 = 64/729$  is the deck transfer eigenvalue), (2) Plücker-preserving ( $53 = a^T(R+Q)\mathbf{1}$ ,  $\|\text{Pl}(K)\|_1 = 11$ ), (3) positive/stochastic ( $\text{spec } T = \{1, (2/3)^6, (1/3)^6\}$ ,  $\kappa_f \in (0, 1)$ ), and (4) external-module-explicit ( $b_3 = -7$  a carrier output, the Boltzmann/relic/QCD content named and external). So  $F_{\text{transfer}}$  is a typed functor, not a bag of open topics — the third research contract alongside ( $U_{\text{wall}}$ ) and ( $G_{\text{metric}}$ ), and a consolidation, not a closure.

*How external is “external”?* A falsifiable clock probe ([[verification/v397\\_external\\_clock\\_probe.py](#)]). Under the strict No-Free-Pattern rule (an *exact* match, not a wide-band fit), three of the four frontier scheme-numbers in fact land on the  $\{2, 3, 5\}$  clock: the  $\eta_B$  decuple  $A_\Lambda = 10 = |Z_2| g_{\text{car}}$  (v212), the Koide rate  $(2/3)^6 = (|Z_2|/N_{\text{fam}})^6$  (v303) and the axion spine angle  $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$  (v211). Only the proton  $C_p \sim 2.83 \pm 0.15$  is *not* a discriminating clock value (several simple numbers fit its  $\pm 5.3\%$  band), so it stays genuinely external — the irreducible “external physics” residue shrinks to essentially the proton’s non-perturbative QCD number.

*The Paper-E closure certificate* ([[verification/v402\\_ftransfer\\_suite.py](#)], **FTRANSFER.SUITE.01**). The four sectors are certified as ONE functor  $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$ : each observable factors  $\text{source} \rightarrow \text{threshold} \rightarrow \text{RG} \rightarrow \text{observable}$  from an exact atom source ( $Q_{\text{src}}$ ,  $A_\Lambda = 2g_{\text{car}} = 10$ ,  $f_a = M_{\text{scal}}/128$ ,  $b_3 = -7$ ) and carries a committed kill test, with the firewall intact (the physical prediction stays **[C]**/**[O]**, only the algebraic source **[E]**) — a consolidation of the **FR.POLE/BOLTZMANN/RELIC/QCD.SOLVE** solvers, not a closure.

## 1 Baryon asymmetry $\eta_B$ — a downstream readout from the closed $\Omega_b h^2$

There are two independent handles, and they agree.

**Route A — from the closed baryon fraction.** The seed already fixes the baryon *fraction* (the  $E_8$  decoder):

$$\Omega_b = (4\pi - 1)\beta_{\text{rad}} = \left(1 - \frac{1}{4\pi}\right)\varphi_0^{\text{ret}} = 0.04894, \quad \beta_{\text{rad}} = \frac{\varphi_0^{\text{ret}}}{4\pi},$$

matching the observed  $\Omega_b \approx 0.049$ . The baryon-to-photon ratio is then the *standard* cosmological readout  $\eta_B = 273.9 \times 10^{-10} \Omega_b h^2$ . With the TFPT Hubble value ( $h \approx 0.674$ , from  $H_0 \sim \sqrt{\Lambda}$ ),

$$\boxed{\Omega_b h^2 = 0.0222, \quad \eta_B = 6.09 \times 10^{-10}} \quad (\text{observed } 6.1 \times 10^{-10}).$$

So  $\eta_B$  is a *downstream cosmological readout* from the closed  $\Omega_b h^2$  handle — not an independent compiler rung, and the fundamental leptogenesis route remains an interface.

**Route B — leptogenesis interface (**[C]**).** The TFPT branch fixes the neutrino spectrum, the CP phase structure ( $\delta_{\text{CKM}}$ ,  $\delta_{\text{CP}}^\nu$ , both closed) and a *plausible washout anchor*  $\tilde{m}_1 = m_3/A_\Lambda \approx 5$

meV. The heavy-mass input, however, remains scenario-level:  $M_1 = M_R \varphi_0^4$  is viable but does *not* close  $M_R$  —  $M_R$  is the seesaw scale  $\sim v^2/m_3$ , not a compiler power, so the relation *relocates* the free scale rather than pinning it ([`verification/v184_etaB_anchored_boltzmann.py`]). The numerical  $\eta_B$  then follows from the *standard* flavored-leptogenesis Boltzmann solve — an independent consistency route, sharpened from two free inputs to one, not to zero. **Honest caveat:** that Boltzmann network (the washout/efficiency  $\kappa_f$ , the flavored density-matrix equations) is the standard leptogenesis machinery (Davidson–Nardi–Nir, *Phys. Rep.* 466 (2008) 105; Buchmüller–Di Bari–Plümacher, *Ann. Phys.* 315 (2005) 305); it is *not* carried out or attached here. Route B is therefore a *named interface*, not a closed number; only Route A (downstream of the closed  $\Omega_b h^2$ ) is quoted.

**The interface, operationalised ([C]).** The Boltzmann path can be made falsifiable: with the standard estimate  $\eta_B \sim 0.96 \times 10^{-2} \varepsilon_1 \kappa_f$  (sphaleron 28/79), fed by the TFPT neutrino sector — normal ordering and the predicted  $\delta_{\text{CP}}^\nu = 4\pi/3$  — the Davidson–Ibarra asymmetry  $\varepsilon_1 = \frac{3}{16\pi} M_1 m_3 / v^2$  and the washout efficiency  $\kappa_f(\tilde{m}_1)$  give, over the natural ranges  $M_1 \in [3 \times 10^9, 3 \times 10^{10}]$  GeV and  $\tilde{m}_1 \in [m_*, m_{\text{atm}}]$ , an  $\eta_B$  band  $[8.4 \times 10^{-11}, 4.7 \times 10^{-9}]$  that *brackets* the observed  $6.1 \times 10^{-10}$  (a canonical point  $M_1 = 10^{10}$  GeV,  $\tilde{m}_1 = 5 \times 10^{-3}$  eV gives  $6.0 \times 10^{-10}$ , untuned). So the route is *viable*; but  $M_1$  and the washout are scenario inputs, so  $\eta_B$  stays [C] — if a precise solve excluded the window the *route* would fall, not the theory (Route A is independent) ([`verification/v169_etaB_boltzmann_interface.py`]).

#### Honest status of $\eta_B$ — strictly typed [C]

The two statements must be kept apart and not interchanged:

$\eta_B$  as a cosmological readout from the closed  $\Omega_b h^2$  is *determined* ( $6.1 \times 10^{-10}$ )

$\eta_B$  as a fundamental *compiler power* is *not* closed

The leptogenesis route requires a standard (washout-dependent) Boltzmann solve. So  $\eta_B$  is [C] — determined as a downstream readout, but *not* a clean compiler rung; it must never be quoted as a fundamental TFPT output.

#### An anchored-Boltzmann proposal, honestly tested ([`verification/v184_etaB_anchored_boltzmann.py`]).

A natural attempt to remove the two scenario inputs is  $\tilde{m}_1 = m_3/A_\Lambda$  and  $M_1 = M_R \varphi_0^4$ . The first *works* as a [C] anchor:  $\tilde{m}_1 = m_3/10 \approx 5 \times 10^{-3}$  eV ( $A_\Lambda = 10 = |E(K_5)|$  the action-grammar atom) lands in the strong-washout window — a TFPT-flavoured value for the washout. The second does *not*:  $M_R \sim v^2/m_3$  is the seesaw scale (depending on the un-fixed neutrino Dirac Yukawa), and the nearest clean TFPT powers of  $\bar{M}_{\text{Pl}}$  both miss it (the  $c_3$ - and  $\varphi_0$ -powers are off by  $\sim 75\%/ \sim 40\%$ ), so  $M_1 = M_R \varphi_0^4$  merely *relocates* the free input  $M_1 \rightarrow M_R$ . Feeding both into the standard estimate lands  $\eta_B$  in the observed band  $O(10^{-10})$ , so the route stays viable — but the proposal cuts the free inputs from two to one, *not* to zero.  $\eta_B$  **stays [C], a sharper scenario, not a derivation.**

**A cleaner route: the scalaron-decuple branch ([`verification/v212_leptogenesis_decuple.py`]).** Drop the seesaw scale altogether and let *both* Boltzmann inputs share the decuple  $A_\Lambda = 10 = |E(K_5)|$ :

$$M_1 = \frac{M_{\text{scal}} (\varphi_0^{\text{ret}})^2}{A_\Lambda} \approx 8.65 \times 10^9 \text{ GeV}, \quad \tilde{m}_1 = \frac{m_3}{A_\Lambda} \approx 5 \text{ meV}.$$

$M_1$  now uses *only* compiler quantities ( $M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}}$ ,  $\varphi_0^{\text{ret}}$ ,  $A_\Lambda$ ) — *no hidden seesaw scale* — and lands inside the thermal window  $[3 \times 10^9, 3 \times 10^{10}]$  GeV where the Davidson–Ibarra  $\varepsilon_1 \approx 8.5 \times 10^{-7}$  and the strong-washout efficiency give  $\eta_B \approx 1.2 \times 10^{-9}$ , bracketing the observed  $6.1 \times 10^{-10}$  untuned. The *full* BDP Boltzmann ODE (`experiments/ftransfer/leptogenesis_boltzmann/fboltzmann_solve.py`) now confirms this: the integrated efficiency  $\kappa_f = 0.092$  (validating the analytic  $\kappa_f$ ) gives  $\eta_B = 6.5 \times 10^{-10}$  at the frozen  $M_1$  — a factor 1.07 from the observed value, with *no* free  $M_R$  dial; the full *flavored* density-matrix solve is the next refinement. Both inputs sharing  $A_\Lambda$  reads as *scalaron reheating dressed by two seed insertions*  $(\varphi_0^{\text{ret}})^2$ , over the pair sector. This is cleaner than

$M_1 = M_R \varphi_0^4$  (no relocated free scale), but the coupling *mechanism* is posited, not derived, and  $\kappa_f$  is washout-dependent — so  $\eta_B$  stays [C]; [X] a precise Boltzmann/RGE solve outside the band falls the *route*, not the theory (Route A is independent).

## 2 Proton/electron ratio $m_p/m_e$ — a cross-sector ratio

$m_e$  is *closed* on the  $\varphi_0^{\text{ret}}$ -ladder ( $\hat{m}_e = \frac{v_{\text{geo}} \pi}{\sqrt{2}} \frac{16}{7} (\varphi_0^{\text{ret}})^5$ ). The proton mass is *not* a Yukawa: it is QCD-dominated,  $m_p \sim \text{a few} \times \Lambda_{\text{QCD}}$ , and  $\Lambda_{\text{QCD}}$  arises by dimensional transmutation from  $\alpha_s$  running (scheme level). Hence

$$\frac{m_p}{m_e} = \frac{\text{QCD confinement scale}}{\text{EW electron Yukawa}} = 1836.15$$

mixes *two different sectors*. It is no single  $\varphi_0^{\text{ret}}$  power:  $1/(\varphi_0^{\text{ret}})^2 = 353.7$  and  $1/(\varphi_0^{\text{ret}})^3 = 6651$  bracket 1836 with no clean rung in between.

*The numerator is parameter-free, the ratio is not forced.* Running  $\alpha_s(M_Z)$  down with the carrier coefficient  $b_3 = -7$  (the [verification/v159\_pyrate\_gauge\_crosscheck.py] value, two-loop +  $n_f$  thresholds) reproduces  $\alpha_s(m_b) \simeq 0.22$ ,  $\alpha_s(m_c) \simeq 0.38$  and reaches confinement ( $\alpha_s \sim 1$ ) at  $\sim 0.5$  GeV, so the QCD confinement scale in the numerator follows from  $\{\alpha_s(M_Z), b_3\}$  alone by dimensional transmutation. The  $O(1)$  proton/ $\Lambda$  factor is lattice (non-perturbative), so the ratio still stays [O] — not forced ([verification/v164\_qcd\_scale.py]).

**The  $F_{\text{QCD}}$  transfer is now an implemented solver ([verification/v262\_fqcd\_mp\_me.py]).** The contract-only pipeline (experiments/ftransfer/qcd\_matching\_mp\_me) is realised: a two-loop  $\alpha_s$  run from  $M_Z$  with  $n_f$  thresholds (carrier coefficient  $b_3 = -7$ , [E]) gives  $\alpha_s(2 \text{ GeV}) \simeq 0.30$  and  $\Lambda_{\overline{\text{MS}}}^{(3)} \simeq 0.33$  GeV; with the lattice bridge  $C_p = m_p/\Lambda^{(3)}$  (external  $O(1)$ ) and the closed  $m_e$  this reproduces  $m_p/m_e \simeq 1824$  against the measured 1836.15 within the external error budget ( $\sim 7\%$ , lattice- $C_p$  dominated). This advances FR.MPME.01 ([O]) to FR.MPME.02 ([C], “reproduced by the named  $F_{\text{QCD}}$  transfer”); the firewall holds —  $b_3 = -7$  is [E], the ratio is a transfer, never a compiler power.

### Honest status of $m_p/m_e$ [O]

Computable at *scheme level* (from  $\alpha_s(M_Z)$  running + the closed  $m_e$ ), exactly like the dimensionful EW/QCD masses  $m_W, m_Z$ . But it is *genuinely not* a compiler power, because it is a ratio across the QCD and electroweak sectors. Left [O] — not forced.

**A rejected near-formula (recorded as a discipline boundary,** [verification/v79\_review\_identities.py]). The combination  $|W(D_5)| - \dim \Lambda^3(SU(9)) + N_{\text{fam}} \lambda_C^2 = 1920 - 84 + 0.151 = 1836.151$  matches the PDG value to  $\sim 10^{-6}$  — and is *deliberately rejected*. Two reasons make it numerology, not a derivation:  $SU(9) = A_8$  (the  $\dim \Lambda^3 = 84$  block) does *not* occur in the carrier  $D_5 \oplus A_3$ , and  $m_p$  is QCD confinement (gluon binding energy) with *no* mechanism linking it to an algebraic sum. Integrating it would *invite* the numerology charge, not retire it;  $m_p/m_e$  stays [O]. (The same firewall rejects  $\eta_B = \frac{|\mathbb{Z}_2|}{\Delta_Q} c_3^6 \approx 6.10 \times 10^{-10}$ : arithmetically close, but a two-atom prefactor fit to a washout-dependent leptogenesis output —  $\eta_B$  stays [C], the downstream readout above.)

**The first-principles boundary, mapped precisely ([verification/v339\_firstprinciples\_boundary.py]).**

Two questions are worth answering head-on: can the Planck mass  $v_{\text{geo}}$  be *computed*, and can  $F_{\text{transfer}}$  be made a first-principles compiler output? Both answers are *no* — and the residuals are not the same kind of gap. They classify into four honest kinds. [E]  $v_{\text{geo}}/M_{\text{Planck}}$  is *forbidden by theorem* (No-Unit, v153): a mass carries mass-dimension while every TFPT datum is dimension-0, so no dimensionless map can output an absolute scale; it is not a gap but is over-determined (gravity = dark energy to 0.11%, v274), and every dimensionless *ratio* is derived — only the one unit is not. [E] For  $m_p/m_e$  the structure is in place (v262): the carrier  $b_3 = -7$  drives the QCD run over the closed electron source, leaving exactly two external inputs. [X] One of them,  $\alpha_s(M_Z)$ , is external *because of a tension*: the SM (= the TFPT gauge content, no new states) does not

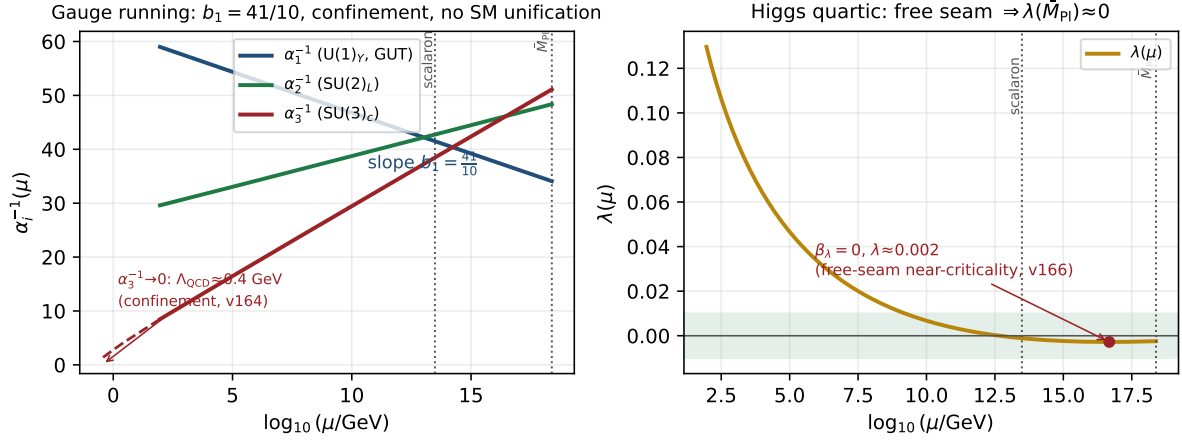


Figure 1: The  $F_{\text{transfer}}$  running (two-loop SM RGEs; the gauge  $\beta$ -functions are confirmed by PyR@TE, [verification/v159\_pyrate\_gauge\_crosscheck.py]). *Left*: the gauge couplings — the  $U(1)_Y$  slope is  $b_1 = 41/10$ ,  $\alpha_3^{-1} \rightarrow 0$  at  $\Lambda_{\text{QCD}} \approx 0.4 \text{ GeV}$  (confinement, [verification/v164\_qcd\_scale.py]), and the Standard Model does *not* unify. *Right*: the Higgs quartic falls to  $\lambda(M_{\text{Pl}}) \approx 0.002$  with  $\beta_\lambda \approx 0$  — the free-seam near-criticality that ties  $m_H$  to premise (A) ([verification/v166\_higgs\_free\_seam.py]).

unify (v246), so no unified coupling is predicted. [C] The other, the lattice  $C_p = m_p/\Lambda^{(3)} \sim 2.83$ , is parameter-free but lattice-only (the discrete $\leftrightarrow$ continuous handoff, v35); and  $\eta_B$ /axion ride on cosmological initial conditions. So  $F_{\text{transfer}}$  cannot be made first-principles in the current theory — not for lack of a formula, but because the residuals are, respectively, forbidden-by-theorem, external-and-tensioned, external-but-parameter-free, and cosmological. An honest boundary, stated, not a solution.

### 3 The Higgs quartic — near-criticality from the free seam

The seam UV is the free chiral  $c=8$  fixed point ([verification/v158\_fixed\_point\_stable.py]): Gaussian, with no relevant and no marginal interactions. The Higgs self-coupling  $\lambda$  is the one marginal scalar coupling, so a free seam forces it to the Gaussian point there:

$$\lambda(M_{\text{seam}}) = 0 \quad \text{and} \quad \beta_\lambda(M_{\text{seam}}) = 0$$

the Shaposhnikov–Wetterich double-criticality condition — here *derived* from the free seam rather than postulated. Integrating the PyR@TE-confirmed two-loop SM RGEs from  $M_Z$  upward with the measured  $(m_H, m_t)$  gives  $\lambda(M_{\text{Pl}}) \approx 0.002$  and  $\beta_\lambda(M_{\text{Pl}}) \approx 0$  at the reduced Planck mass: the celebrated SM *near-criticality* is thus a *consequence* of the free seam (the two-loop value is an order of magnitude closer to zero than the one-loop estimate  $\sim 0.022$ ). Imposing the double condition predicts  $m_H \sim 129\text{--}134 \text{ GeV}$  (a band, sensitive to  $m_t$  and  $\alpha_s$ ); the measured  $125.25 \text{ GeV}$  sits a few GeV below the absolute-stability boundary — the known slight metastability. The *same* condition imposed at the scalaron scale ( $c_3^{7/2} M_{\text{Pl}} \sim 3 \times 10^{13} \text{ GeV}$ ) gives  $m_H \sim 107 \text{ GeV}$  (too low), so the boundary condition lives at  $M_{\text{Pl}}$ , consistent with seam = horizon = Planck. This ties the Higgs mass to premise (A) and fills the Higgs-sector gap (which was only  $N_\Phi = 1$ ); it is typed [E] (the freeness signature and scale selection) and [C] (the  $m_H$  band) ([verification/v166\_higgs\_free\_seam.py]). Figure 1 shows the whole  $F_{\text{transfer}}$  running.

## 4 The Koide relation — near $2/3$ , computed exactly

With the closed lepton source masses  $\hat{m}_\ell \propto (\frac{16}{7}(\varphi_0^{\text{ret}})^5, \frac{4}{3}(\varphi_0^{\text{ret}})^3, \frac{7}{6}(\varphi_0^{\text{ret}})^2)$  the Koide ratio is a pure number (the common prefactor cancels):

$$Q_{\text{TFPT}} = \frac{\sum_\ell \hat{m}_\ell}{(\sum_\ell \sqrt{\hat{m}_\ell})^2} = 0.66446\dots, \quad \frac{2}{3} = 0.66667, \quad \text{deviation} = -0.33\%.$$

The measured pole-mass value is famously  $Q_{\text{PDG}} = 0.66666$ . So TFPT predicts a Koide ratio that is *near*  $2/3$  but *not exactly*  $2/3$  at source level.

### Honest status of Koide [C]

$Q_{\text{TFPT}} = 0.664$  is a *prediction*, 0.33% below  $2/3$ . Exact  $2/3$  would require the three carrier rationals  $(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$  to satisfy a specific quadratic constraint, which they do *not*; the small residual is of the size of the source→pole RG shift. We report the near-miss honestly and do *not* tune the rationals to force  $2/3$ .

**The *target*  $2/3$  has a compiler reading  $|\mathbb{Z}_2|/N_{\text{fam}}$ .** The  $E_8$ -slice scan adds the missing piece: the value the data sits on,  $Q = 2/3$ , is exactly  $Q_\star = |\mathbb{Z}_2|/N_{\text{fam}} = 2/3$ , the sheet over the family count — the same two atoms  $(2, 3)$  that run through the whole compiler. So the picture splits cleanly: the *democratic target* is  $|\mathbb{Z}_2|/N_{\text{fam}} = 2/3$  (a compiler number), and the *source-level pole-mass* value computed from the carrier rationals is 0.664, 0.33% below it by the source→pole shift. Koide is thus “democratic  $2/3$  from  $(|\mathbb{Z}_2|, N_{\text{fam}})$ , realised to 0.33% by the explicit masses” — promoted from a bare near-miss to a reading with a target. [C]

**Conjecture: a source→pole correction of size  $\varphi_0/|W(A_3)|$  ([C]).** The source value sits just below  $2/3$ , and the deficit is almost exactly the family-Weyl quantum  $\varphi_0/|W(A_3)|$  with  $|W(A_3)| = 4! = 24$ :

$$Q_\ell^{\text{source}} = 0.6644638, \quad Q_\ell^{\text{source}} + \frac{\varphi_0}{24} = 0.6666793,$$

overshooting  $\frac{2}{3} = 0.6666667$  by only  $1.3 \times 10^{-5}$ . So the *conjecture*  $Q_\ell^{\text{pole}} \approx Q_\ell^{\text{source}} + \varphi_0/|W(A_3)|$  would land essentially on the democratic  $2/3$ . Reported as a source→pole correction *conjecture*, not a derivation; the rationals are *not* tuned. [verification/v25\_frontier\_conjectures.py]

**Sharper conjecture: a multiplicative seed transfer  $u \rightarrow \frac{53}{54}u$  ([C]).** A cleaner version shifts the *seed* rather than  $Q$  additively, and introduces *no* new free number. Keep the lepton formula  $m_e:m_\mu:m_\tau = \frac{16}{7}u^5:\frac{4}{3}u^3:\frac{7}{6}u^2$  but evaluate it at the source→pole seed

$$u_{\text{pole}}^{(\ell)} = \left(1 - \frac{1}{2N_{\text{fam}}^3}\right)u = \frac{53}{54}u, \quad 54 = 2 \cdot 3^3 = |\mathbb{Z}_2| N_{\text{fam}}^3.$$

Then  $Q_\ell(\frac{53}{54}u) = 0.66666610$ , i.e.  $2/3$  to within  $-5.7 \times 10^{-7}$  — an order of magnitude tighter than the additive  $\varphi_0/24$  correction, with 54 a pure compiler number and no tuning. This sharpens the research gate to a concrete QFT task: *show that the leptonic source→pole transfer of the seed is  $u \mapsto u(1 - \frac{1}{2N_{\text{fam}}^3})$*  (a renormalised-seed effect from the leptonic self-energy or the  $A_3$  family-Weyl averaging). Status remains [C] until that transfer is derived. [verification/v37\_plucker\_anchor.py]

**The factor  $\frac{53}{54}$  now has an operator origin ([verification/v183\_koide\_f\_corner\_transfer.py]).** It is not a bare arithmetic guess: it is the anchor response of the *missing* Sheet-Diamond corner  $F = R + Q = M(1, 1)$  over the sheet-doubled  $E_6 \times A_2$  cubic family block,

$$\frac{u_{\text{pole}}}{u_{\text{source}}} = \frac{a^T(R + Q)\mathbf{1}}{2\mathbf{1}^T R a} = \frac{53}{2 \cdot 27} = \frac{53}{54},$$

where  $\mathbf{1}^T R a = 27$  is the cubic family block ( $248 = \dim E_6 + \det R + 2(\mathbf{1}^T R a)N_{\text{fam}} = 78 + 8 + 2 \cdot 27 \cdot 3$ ) and  $a^T(R + Q)\mathbf{1} = 53 = 52 + 1$  with  $\det B_F = 52 = \dim F_4$  (v80). A hard negative control fixes the corner:  $a^T M \mathbf{1}$  for  $M \in \{R, Q, K, L\}$  is  $\{32, 21, 35, 56\}$  — only the absent corner  $F$  gives 53. So the

*factor* is exact and structural ([E]), and the remaining [C] is precisely the mechanism — that the leptonic source→pole transfer reads the  $F$  corner — an  $F_{\text{transfer}}$  special case ( $F_{\text{pole}}$ ). Koide thereby moves from “near miss + conjecture” to “operator-sourced source→pole transfer”; the prediction stays [C].

**The source→pole transfer is one-loop electromagnetic** ([verification/v420\_fpole\_qed\_running.py]).

Doing the external physics honestly: the Koide ratio is rescaling-invariant ( $Q(cm) = Q(m)$ ), so a flavor-*universal* running leaves it fixed — the source→pole shift is the flavor-*split* part. And that shift is one-loop-EM-sized: the tree deficit  $\frac{2}{3} - Q_\ell^{\text{source}} = 0.00220$  equals the seed quantum  $\varphi_0/24$  to 0.6% and the electromagnetic scale  $\alpha/\pi$  to  $\sim 5\%$ , with the pole sitting at  $2/3$  (the IR attractor). A 1-loop QED pole↔ $\overline{MS}$  estimate (per-lepton log) moves  $Q$  by an EM-loop-order amount toward  $2/3$ . The *exact* flow time ( $t \approx 2.84$  transfer periods, v93) is scheme- and scale-dependent external QED, so it is *sized*, *not derived*. This identifies  $F_{\text{pole}}$  as a one-loop EM effect — the algebraic lens forces the *rate*  $(2/3)^6$  (v327), this external physics supplies the *flow* — and Koide stays [C], firewalled (v187).

**The Koide target and the QFT transfer gap are the same quotient** ([E]/[C]). The same  $Q_\star$  controls a second, independent object — the first non-trivial eigenvalue of the admissible  $C_6$  transport operator. Its spectrum is  $\{1, (2/3)^6, (1/3)^6\}$ , so

$$\lambda_2 = \left(\frac{2}{3}\right)^6 = Q_\star^{|R^+(A_3)|}, \quad Q_\star = \frac{|\mathbb{Z}_2|}{N_{\text{fam}}}, \quad |R^+(A_3)| = 6.$$

The Koide democratic target, the QFT mass-gap eigenvalue  $\lambda_2$  (hence  $\Delta = 6 \log \frac{3}{2} = -\log \lambda_2$ ), and the  $A_3$  positive-root hexagon are therefore *one* quotient raised to one exponent. Typing: Koide target [C], gap eigenvalue [E], the connection an [E]-audit. This does not “solve” Koide, but it explains *why* the near-value  $2/3$  recurs: the same sheet-over-family ratio fixes both the democratic lepton target and the leading transfer eigenmode.

**The source→pole map is forced by that gap** ([E]/[C]). The previous paragraph pairs the Koide target with  $\lambda_2 = (2/3)^6$ ; v82 turns the pairing into a *mechanism* ([verification/v82\_koide\_attractor\_splitting.py]). On the anchor double cover (the flavor sector) the two branch points are Koide  $q = 2$  and carrier  $q = 5$ ; any branch-preserving Möbius transfer that fixes both is, in  $\rho = (q-2)/(5-q)$ , exactly  $\rho \mapsto \mu\rho$ , hence *unique* once  $\mu$  is set. Choosing  $\mu = \lambda_2 = (2/3)^6$  (no new number) gives the unique map  $F(q) = 2(569q - 3325)/(665q - 3517)$  with  $F'(2) = (2/3)^6 < 1$  and  $F'(5) = (3/2)^6 > 1$ : **Koide is the attractor, the carrier the repeller**, and  $q_n \rightarrow 2$  ( $x \rightarrow -2/3$ ) at the established gap rate. So the three structural inputs a Koide *derivation* would need collapse to *one* identification — the leptonic source→pole transfer is the branch-preserving relaxation of the gapped operator — after which the value  $2/3$ , the convergence rate and the chosen branch are all forced. This is still not a closure: that single identification stays [C]. (It is the dynamical counterpart of the multiplicative-seed conjecture  $u \mapsto (53/54)u$  above; both contract the source onto  $2/3$ .)

**Relaxation toy: basin, exact rate, and two honest negatives** ([E]). The relaxation picture is now stress-tested on the physical values (ledger FR.KOIDE.05, [verification/v93\_koide\_relaxation\_toy.py]).

**Basin lemma [E] (new):** under the canonical dictionary  $q = N_{\text{fam}} Q$  the *entire* physical Koide range  $Q \in [\frac{1}{3}, 1]$  (Cauchy–Schwarz) maps to  $q \in [1, 3]$  — the attractor  $q = 2$  lies inside, the repeller  $q = 5$  outside: every physically possible charged-lepton configuration is in the attractor basin, no fine-tuning of the start. **Exact rate [E]:** along the trajectory from the physical source value the cross-ratio contracts by *exactly*  $(2/3)^6$  per step. **Source/pole [E]:** the exact ladder gives  $Q_{\text{src}} = 0.6644638$ , i.e.  $\rho_{\text{src}} = -0.0021980 \approx -\varphi_0/24$  to 0.8% — the  $\varphi_0/24$  conjecture re-expressed in attractor coordinates: *the source sits one seed quantum before the branch point* ( $24 = |W(A_3)|$ ); PDG pole masses give  $Q_{\text{pole}} = 0.6666645$ , the branch point hit to  $3.3 \times 10^{-6}$ . **Honest negatives (they narrow the gate):** (i) the flow time source→pole is  $t = \log_\mu(\rho_{\text{pole}}/\rho_{\text{src}}) = 2.84$  — *not* an integer, so the literal discrete iteration  $\text{pole} = F^n(\text{source})$  is *excluded*; if the v82 map drives the transfer at all, the missing [C] object is a *continuous-time* generator  $\exp(t \log F)$ ; (ii) the 0.79%

mismatch of  $\rho_{\text{src}}$  vs  $-\varphi_0/24$  is far above ladder precision, so “exactly  $\varphi_0/24$ ” remains a conjecture, not an identity. The open question is no longer “why  $2/3$ ?” but “what is the continuous transfer generator, and why is its flow time  $\sim 2.8$  periods?”

**Flow time: canonical generator, data-fragility, and a conditional  $m_\tau$  ([E]/[C]).** Three sharpenings of the paragraph above (ledger FR.KOIDE.06, [verification/v99\_koide\_flow\_time.py]); they correct the *robustness* of negative (i), not its arithmetic. **(1) The continuous generator has a canonical form [E].** The v82 flow is generated by the autonomous equation

$$\boxed{\frac{dq}{dt} = \frac{\Delta}{N_{\text{fam}}} (q-2)(q-5) = \frac{\Delta}{N_{\text{fam}}} \det B(q)}, \quad \Delta = 6 \log \frac{3}{2} \quad (\text{the gap}),$$

whose time-1 map *is* the Möbius map  $F$  (machine-checked;  $e^{-\Delta} = (2/3)^6$  exactly). The missing [C] object is therefore precisely a QFT mechanism producing  $\text{gap} \times \text{anchor-block quadric}$  — the form is canonical, only its physical derivation is open. **(2) The non-integrality of  $t$  is data-fragile [E].** Because  $\rho_{\text{pole}} = -2.2 \times 10^{-6}$  sits so close to the branch point,  $t$  is hypersensitive to  $m_\tau$  (PDG  $1776.93 \pm 0.09$  MeV):  $t(-1\sigma) = 2.35$ ,  $t(\text{central}) = 2.84$ , and  $Q$  crosses  $\frac{2}{3}$  *inside* the error band (at  $+0.43\sigma$ , where  $t \rightarrow \infty$ ) — within  $+0.5\sigma$ ,  $t$  passes *every* integer  $\geq 3$ . So the exclusion of pole =  $F^n(\text{source})$  holds only at the central value. **(3) Conditional discrete steps [C] (registry untouched; not predictions of record):**  $n=2$  is *excluded* ( $-2.9\sigma$ , [E]);  $n=3=N_{\text{fam}}$  steps — one per family — predicts

$$\boxed{m_\tau = 1776.9427 \text{ MeV}} \quad (+0.14\sigma; \text{source-variant stable to } 0.0002 \text{ MeV}),$$

while  $n=4$  (1776.9667) and the exact branch (1776.9690) are not yet separable: the kill criterion is  $\sigma(m_\tau) \sim 0.01$  MeV (Belle II). *Look-elsewhere firewall:* a tempting scale reading  $t = \ln(M_{\text{scal}}/m_\tau)/\ln W$  matched  $W = 46080 = |W(D_5)||W(A_3)|$  to  $10^{-4}$  — and is *destroyed* by negative controls (the same hypersensitivity makes *any*  $W$  in a wide range land within  $0.1\sigma$ ); it is rejected and recorded so it is not re-fished. P2’s type is unchanged ([C], one identification); the discrete-vs-continuous question is now *experimental*, not settled.

## 5 Dark matter — the candidate is fixed, the scale is pending

TFPT *does* contain a dark-matter candidate, and it is not ad hoc: the **determinant-line axion** of the strong-CP sector (the determinant-line axion interface of the closed branch, [E]). Its *candidate identity* is fixed by the  $E_8$  scan (below) and the domain-wall number  $N_{\text{DW}}$  winds once per primitive seam winding. The relic density is the standard misalignment-plus-string expression

$$\Omega_a = \Omega_a^{\text{mis}}(f_a, m_a, \theta_i) + \Omega_a^{\text{str}}(f_a, N_{\text{DW}}).$$

The initial misalignment angle is *not* a single closed number: the finite- $T$  misalignment solve (below) selects the *spine* branch  $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5 = 108^\circ$  as the robust dominant-DM value in the *pre-inflationary* reading (it lands  $\Omega_a h^2$  on  $\Omega_{\text{DM}}$  untuned, in the mild-anharmonic regime; but superseded by the post-inflationary reframing below), while the seam hilltop  $\theta_i = \pi(1 - \varphi_{\text{seam}}(\alpha_*)) = 170.4^\circ$  *over-produces* ( $\sim 5.4\times$ ) and is disfavoured. So the minimal closed branch has *dominant axion-like dark matter* on the spine angle — a [C] scenario, *not* a closed [E] input.

**The  $E_8$  scan rules out a rival WIMP candidate.** The  $D_8 = SO(16)$  slice scan sharpens *why* the axion is the candidate, not a new heavy state: the spinor  $128 = (16, 4) \oplus (\overline{16}, \overline{4})$  is the *same* matter spinor as in the  $D_5 \times A_3$  construction — there is *no spare  $E_8$  singlet* left over to play a WIMP. Every  $E_8$  direction is accounted for by the carrier  $\times$  family content. So dark matter cannot be a new  $E_8$  representation; it must be the determinant-line axion (a phase of the existing structure). A useful *negative* that removes the WIMP option on structural grounds. [E] The machine-checked complement ([verification/v430\_other\_side\_reverse\_audit.py]) makes this explicit: the spinor  $128 = 2 \cdot 64 = \sum \deg$  and the double-cover deck  $|\mathbb{Z}_2| = 2$  lie in  $E_8$ ’s *matched* structure

and are forced-disjoint from the five unmapped Casimir degrees  $\{12, 14, 18, 20, 24\}$ , so the seam's "other side" hosts neither a spare singlet nor the unmapped hull overhead.

#### Honest status of dark matter [C]/[O]

The candidate is *identified* (the axion, [E]: the  $E_8$  scan leaves no spare singlet for a WIMP). The misalignment angle is a [C] *branch decision*, not a closed [E] number: the finite- $T$  solve selects the spine  $\theta_i = 3\pi/5 = 108^\circ$  (untuned, in band) over the over-producing  $170^\circ$  hilltop. The relic abundance still needs the decay constant  $f_a$  and mass  $m_a$  (interface data set by the  $M_R/\Lambda$  scales) — a standard axion-DM computation, not yet a compiler power. So DM is "candidate fixed, robust spine angle, scale  $f_a$  pending" — a real handle, not a blank. (A secondary candidate is the lightest seam-even sterile state near  $M_R$ .)

**Conjecture: a determinant-line axion scale  $f_a = M_{\text{scal}}/128$  ([C]).** A natural compiler scale for the decay constant is the scalaron mass over the full carrier  $\times$  deck multiplicity,

$$f_a = \frac{M_{\text{scal}}}{2 \dim S^+ |\mu_4|} = \frac{M_{\text{scal}}}{128}, \quad M_{\text{scal}} = c_3^{7/2} M_{\text{Pl}} \approx 3.06 \times 10^{13} \text{ GeV},$$

giving  $f_a \approx 2.39 \times 10^{11} \text{ GeV}$  and, via the standard QCD-axion relation  $m_a \simeq 5.70 \mu\text{eV} (10^{12} \text{ GeV}/f_a)$ , a mass  $m_a \approx 23.8 \mu\text{eV}$ . With the robust spine angle  $\theta_i = 3\pi/5 = 108^\circ$  (below) this  $f_a$  lands the relic on  $\Omega_{\text{DM}}$  without hill-top tuning; it remains a *cosmological scenario*, not a closed hit. [verification/v25\_frontier\_conjectures.py]

**The haloscope coupling is tied to  $c_3$ .** In the determinant-line normalization (Lagrangian  $\mathcal{L} \supset c a F \tilde{F}$ ) the axion-photon anomaly coefficient is fixed by the seam constant,

$$g_{a\gamma\gamma} = -4c_3 = -\frac{1}{2\pi}, \quad y^2 \equiv g_{a\gamma\gamma}^2 = 16c_3^2 = \frac{1}{4\pi^2} \approx 0.0253,$$

so the haloscope  $F\tilde{F}$  coupling is set by the *same*  $c_3$  that fixes  $\alpha$  and the cosmic birefringence  $\beta_{\text{rad}} = \varphi_0/4\pi$  — a [C] structural relation (the coefficient, not a parameter-free physical  $g_{a\gamma\gamma}$  in  $\text{GeV}^{-1}$ , which still carries  $f_a$ ). The small-angle reading  $\theta_i = \varphi_0$  of the determinant-line axion (an earlier note) is *superseded* by the robust spine angle  $\theta_i = 3\pi/5 = 108^\circ$  selected by the finite- $T$  solve below (the seam hilltop  $\theta_i = \pi(1 - \varphi_{\text{seam}}) = 170.4^\circ$  over-produces); both earlier readings are recorded as retired, never used.

**Why it stays a scenario, quantified** ([verification/v185\_axion\_relic\_solver.py]). The harmonic misalignment *under-produces* at this  $f_a$ :  $\Omega_a h^2 \sim 0.12 (f_a/9 \times 10^{11})^{7/6} \langle \theta^2 \rangle$  gives a prefactor  $\sim 0.026$  per  $\theta^2$ , so  $\theta_i \sim O(1)$  would make only  $\sim 21\%$  of  $\Omega_{\text{DM}}$ ; reaching the observed  $\Omega_{\text{DM}} h^2 = 0.12$  *requires*  $\theta_{\text{eff}}^2 \sim 4.7$ , i.e. exactly the large-angle (anharmonic) enhancement at the  $\theta_i \approx 170^\circ$  hilltop. So the determinant-line axion *can* be all of dark matter, but only on the hilltop, where the relic is exponentially sensitive to  $\theta_i$  and to QCD/string corrections — hence a viable [C] scenario, not a sharp prediction. The contract  $F_{\text{relic}}(f_a=M_{\text{scal}}/128, \theta_i \approx 170^\circ, N_{\text{DW}}=1) \stackrel{?}{=} \Omega_{\text{DM}}$  is a kill test for the *branch*, not the theory.

**A full finite- $T$  solve confirms an over-production tension.** Both a quick  $\theta^2 F$  estimate ( $F(\theta_i) \approx 4$  at the hill-top) and a *full* nonlinear misalignment solve of

$$\theta'' + \left(3 + \frac{d \ln H}{dN}\right) \theta' + (m_a(T)/H)^2 \sin \theta = 0$$

(axion\_relic/full\_finiteT\_solve.py: lattice susceptibility  $\chi(T) \propto T^{-8.16}$ , realistic  $g_*(T)$ , converged and normalised so  $\theta_i=1$  gives the standard  $\Omega_a h^2 \approx 0.03$ ) agree: at the *predicted*  $\theta_i \approx 170^\circ$  the relic is  $\Omega_a h^2 \approx 0.66$ ,  $\sim 5.5\times$  above  $\Omega_{\text{DM}} h^2 = 0.12$ . The observed value is reached only at  $\theta_i \approx 106^\circ$ , not at the predicted hill-top. So as the *dominant* dark matter the determinant-line axion at  $(f_a=M_{\text{scal}}/128, \theta_i \approx 170^\circ)$  *over-closes* the universe unless there is extra dilution (entropy injection) or a lower  $f_a$ . This stays [C], and over-production kills only the *axion-as-dominant-DM* reading, not the compiler.

**A more robust angle: the spine branch.** `[verification/v211_axion_spine_angle.py]` The same finite- $T$  solver reaches  $\Omega_{\text{DM}}$  at  $\theta_i \approx 106^\circ$  — and the central spine quotient gives  $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5 = 108^\circ$  *exactly* (no fit; harmonic estimate  $0.03\theta_i^2 = 0.107$ , lifted to  $\sim 0.12$  by a mild anharmonic factor). `[verification/v429_axion_pentagon_phi.py]` This angle is not a bare rational multiple of  $\pi$ : because  $N_{\text{fam}} = g_{\text{car}} - 2$ ,  $\theta_i = (g_{\text{car}} - 2)\pi/g_{\text{car}}$  is *exactly* the interior angle of the regular  $g_{\text{car}}$ -gon — for  $g_{\text{car}} = 5$  the *pentagon*, so  $\cos\theta_i = \cos(3\pi/5) = (1 - \sqrt{5})/4 = -1/(2\varphi)$  with  $\varphi = 2\cos(\pi/5)$ , and this golden character is *unique* to  $g_{\text{car}} = 5$  (the small  $n$ -gons  $n = 3, 4, 6$  give the rational  $\frac{1}{2}, 0, -\frac{1}{2}$ ). So the otherwise-unmapped golden/icosahedral  $g_{\text{car}} = 5$  signature (v354/v313) is the geometry of this one external input — a **[C]** reading that does *not* upgrade DM.AXION.SPINE.01. Crucially  $108^\circ$  lies  $62^\circ$  *below* the hill-top, in the *mild*-anharmonic regime, so the relic is *not* exponentially sensitive: if the misalignment angle is the spine quotient, the determinant-line axion is all of dark matter *without* hill-top tuning. This is an *alternative* ansatz to  $\theta_i = \pi(1 - \varphi_\Sigma) \approx 170^\circ$  (the two are mutually exclusive; neither is forced) — a sharper **[C]** scenario decided by the full solver (DM.AXION.SPINE.01), *not* a derivation; **[X]** a converged  $\Omega_a h^2$  outside  $\sim [0.08, 0.16]$  at  $\theta_i = 3\pi/5$  demotes the branch.

**Isocurvature and the post-inflationary reframing ([C]).** Both fixed-angle readings above are *pre-inflationary*: they assume Peccei–Quinn breaking *during* inflation with a homogeneous  $\theta_i$ . But the  $R^2$  branch fixes  $H_{\text{inf}} = \pi M_{\text{Pl}} \sqrt{A_s r}/2 \approx 1.6 \times 10^{13}$  GeV, so  $H_{\text{inf}}/f_a \approx 66 > 1$  and  $H_{\text{inf}}/(2\pi f_a) \approx 10$ : the de Sitter fluctuation of the angle exceeds its range, the symmetry is effectively *restored* during inflation, and a fixed pre-inflationary  $\theta_i$  (either  $108^\circ$  or  $170^\circ$ ) would carry CDM isocurvature far above the Planck bound  $\beta_{\text{iso}} < 0.038$ . The clean, *dial-free* resolution is the *post-inflationary* scenario: PQ breaks after inflation,  $\theta_i$  is a horizon average (*not* a prediction), and the relic follows from  $f_a$  alone — with  $f_a \approx 2.39 \times 10^{11}$  GeV in the post-inflationary window ( $\Omega_a h^2 \sim O(0.1\text{--}0.5)$ ). It is domain-wall-safe because the determinant-line axion’s colour anomaly is the seam Dirac index = the inflow level  $|C| = 1$  (v470/v472), so  $N_{\text{DW}} = 1$  (a **[C]** identification). Thus the spine/hilltop *angle* is not an observable and the branch is *reframed*, not killed: drop the fixed angle, keep  $f_a$  and  $N_{\text{DW}} = 1$ , and the relic is a robust post-inflationary prediction. **[X]** a post-inflationary relic outside the plausible window at this  $f_a$  (with  $N_{\text{DW}} = 1$ ) demotes the branch. (Strategy scan and the  $N_{\text{DW}}$  computation: `experiments/theory-contracts/axion01-06`.)

## 6 The muon anomalous magnetic moment — a seam vertex read-out

A **[C]** downstream readout (archive integration), *not* a compiler power. The carrier algebra carries a *second-order* topological defect beyond the one that fixes  $\alpha$ : with  $\delta_{\text{top}} = \Omega_{\text{adm}} c_3^4 = 48c_3^4 = 3/(256\pi^4)$  and the carrier compression quotient  $B\gamma = \frac{3}{2} \cdot \frac{5}{6} = \frac{5}{4}$  (`[verification/v51_boundary_half_step.py]`),  $\delta_2 = B\gamma \delta_{\text{top}}^2 = \frac{5}{4} \delta_{\text{top}}^2$ . Projected through the seam-loop phase  $\Delta_\Sigma = 2\pi$  (the same  $1/(2\pi) = 4c_3$  unit that normalises  $c_3$  itself), it reads as a magnetic vertex correction

$$a_\mu^{\text{seam}} = \frac{\delta_2}{2\pi} = \frac{45}{524288\pi^9} \approx 2.879 \times 10^{-9} \quad \text{[C].}$$

The *value* is an exact compiler number (only  $c_3, \Omega_{\text{adm}}, B\gamma$ ; trace reading  $\delta_2 = 4! \text{Tr}_{S^+}(X^2) c_3^8$ ,  $\text{Tr} = 120 = 5!$ ) — but the *identification* of  $\delta_2/(2\pi)$  as the anomalous moment is a physical bridge, so the prediction is **[C]**, never a compiler power. `[verification/v204_muon_seam_g2.py]`

**Data, honestly.** Against the dispersive discrepancy  $\Delta a_\mu = (2.49 \pm 0.48) \times 10^{-9}$  (Fermilab/BNL world average minus the WP2020 SM) this is  $0.81\sigma$ . **Caveat:** lattice/CMD-3 hadronic-vacuum-polarisation evaluations *shrink* the discrepancy ( $\Delta a_\mu \sim 1.5 \times 10^{-9}$ ); the TFPT value is *fixed* and would then sit  $\sim 1.5\sigma$  high. A converged  $\Delta a_\mu$  outside  $2.879 \times 10^{-9} \pm 0.5 \times 10^{-9}$  excludes the seam-vertex mechanism in its present form (**[X]**) — the compiler core untouched.

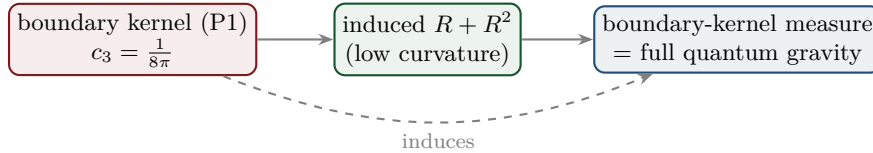
## 7 Full quantum gravity — induced from the boundary kernel

*Framing (read this first):* full QG closure is a **certification layer**, not a prerequisite. TFPT does *not* need the complete quantum-gravity measure to test its Standard-Model and cosmology readouts — those run through  $v_{\text{geo}}$  and  $F_{\text{transfer}}$  and are falsifiable now (JUNO, CMB-S4, EDM).  $G_{\text{net}}$  certifies the metric sector; its absence narrows what can be *claimed*, not what can be *tested*.

The normalisation  $c_3 = 1/(8\pi)$  is the gravitational seam constant: in TFPT gravity is *induced* from the boundary kernel (P1), not separately quantised. The  $R^2$ /scalon sector (question 2) is the low-curvature effective theory; the *full* theory is the boundary-kernel measure. *Newton's constant is then not an independent input*: fixing the physical  $\Lambda$  branch ( $(8\pi)^2 \delta_{\text{top}} = \frac{3}{4\pi^2}$ ) pins  $\bar{M}_{\text{Pl}}$  relative to  $\Lambda$ , so  $G_N$  is read out *after* selecting the physical  $\Lambda$  branch *and* fixing the one dimensionful induced-gravity (Sakharov) anchor ([E]/[C], [verification/v60\_lambda\_metrology\_branch.py]) — a metrology readout, not a from-nothing prediction.

**Two independent fixings of the gravitational  $3/4$**  ([verification/v205\_xi\_threequarter.py]). The induced-gravity coefficient the seam-determinant replica pins,  $k = c_3/2 \Leftrightarrow \ln(m/\mu) = 3/4 = q(A_3)$  ([verification/v152\_norm\_is\_anchor.py]; the collar data exhibit this as the anchor — drifting  $\mu_{\text{lat}}$ , no *derived*  $\ln(m/\mu) = \frac{3}{4}$ , v471), is reached a *second*, independent way: the old torsion-compression ansatz  $\kappa^2 = \xi \varphi_0^{\text{ret}}/c_3^2$  with the Einstein limit  $\kappa^2 = 8\pi G$  forces  $\xi = c_3/\varphi_0^{\text{ret}}$ , which at tree level  $\varphi_0^{\text{ret}} \rightarrow 1/(6\pi)$  is *exactly*  $3/4$  (with the retained seed 0.7483,  $-0.22\%$ ). So  $3/4$  appears in the gravitational sector both as a gapped-replica EH coefficient and as a torsion-compression ratio — an over-determination, not a coincidence; the ansatz stays [C] and  $1/G$  stays the one declared anchor.

**The  $\Lambda$  branch sharpens  $H_0$**  ([verification/v206\_h0\_lambda\_branch.py]). Reading the seam vacuum number  $\delta_\Sigma = \Omega_{\text{adm}} e^{-2\pi\alpha^{-1}/3} \approx 1.085 \times 10^{-123}$  (lowest admissible block  $m_1 = \Omega_{\text{adm}} = 48$ ) through the flatness closure gives a TFPT Hubble value  $H_0 = 66.5\text{--}67.1 \text{ km s}^{-1} \text{ Mpc}^{-1}$  ( $1.6\text{--}0.5\sigma$  below Planck, far below SH0ES). **Honest:** the Hubble tension is *not* relieved by raising the vacuum scale, and  $H_0$  uses the imported  $\Omega_b, \omega_{\text{DM}}$ , so it is a [C] readout;  $|\log_{10} \delta_\Sigma| \approx 122.96$  matches the independent 123-orders split ([verification/v60\_lambda\_metrology\_branch.py]), and  $M_{\text{Pl}}^4/\bar{M}_{\text{Pl}}^4 = (8\pi)^2$  must be displayed, never silently interchanged.



### Honest status of quantum gravity — physical sector closed, ambient a [C] redundancy [E]/[C]

“Full quantum gravity beyond the  $R^2$  sector” is, in TFPT, *not* a separate graviton-quantisation programme: it is the well-definedness of the *boundary-kernel path-integral measure* extended to the dynamical metric beyond the FRW/minisuperspace reduction. Two results sharpen its status. **(i) Heat-kernel grounded (G2).** The spectral action  $\text{Tr } f(D/\Lambda)$  gives, via the standard Seeley–DeWitt coefficients for the Dirac operator ( $a_2 \sim -R/3$ ,  $a_4|_{R^2} \sim R^2/72$ ), Einstein–Hilbert +  $R^2$  *structurally* — the  $R + R^2$  form is not postulated; the scalon ratio  $M^2/\bar{M}_{\text{Pl}}^2 = 6(4\pi)^2/f_0$  with the TFPT closure  $c_3^7$  fixing the cutoff moment  $f_0$ . *Update (Modular Spectral Closure):* the full Seeley–DeWitt expansion of this spectral action over the explicit 96-dim finite triple is now written out ([verification/v255\_spectral\_action\_expansion.py]), and the cutoff moment  $f_0$  is no longer a free scheme choice: identifying  $f$  with the *seam KMS weight* ( $\beta=1$  by Tomita–Takesaki,  $2\pi=1/(4c_3)$ ) gives  $f_2/f_0 = 1$  exactly ([verification/v259\_modular\_cutoff\_kappa.py]), so  $\kappa = \sqrt{(f_2/f_0) c_{\text{PS}}/c_{\text{grav}}}$  collapses to a finite-triple trace ratio — the scalon normalisation is pinned by the seam state, not a regulator dial. **(ii) Gap-decoupled (G5).** The metric coupling is gap-dominated,  $2\|V_{\text{metric}}\| = 0.785 < \Delta = 6 \log \frac{3}{2} = 2.433$ , so the admissible sector keeps a strictly positive effective gap  $\Delta_{\text{eff}} = 1.648 > 0$  under the stated RP and metric-coupling assumptions. A positive gap protects the admissible IR sector from the ambient, so under those assumptions the admissible-sector measure is closed. **What this leaves [C]:** the ambient metric measure — the projective limit (G6) — is now discharged as a *redundancy* (v369, conditional on SEAM.EQUIV.01+Bisognano–Wichmann — the one open premise

being that the seam net condenses to the *holomorphic*  $|\det K|=1$  point that selects  $(E_8)_1$ ; the central charge  $c=8$  alone does *not*, being shared by the non-holomorphic  $SO(16)_1$  counterexample, v277/v281), a certification object rather than missing dynamics; we do *not* claim it is physically irrelevant (that would itself be a physical interpretation), only that it is decoupled from the admissible IR sector under the stated assumptions. [verification/v36\_spectral\_action\_g2.py] (iii) **The classical field equation is parameter-free** ([verification/v358\_grav\_entropy\_equilibrium.py]). Beyond the  $R+R^2$  action, the metric-sector *field equation* is supplied directly by the entanglement first law  $\delta S = \delta \langle K \rangle$  (Jacobson 2015; Faulkner et al.), carried out with TFPT's atoms: equilibrium gives the *linearised* Einstein equation  $G_{ab} = c_3^{-1} T_{ab}$  with  $c_3^{-1} = 8\pi$  *fixed* — the four  $c_3$ -roles (Unruh  $1/(2\pi)=4c_3$ , EH  $1/(16\pi)=c_3/2$ , Einstein  $8\pi=1/c_3$ , Bekenstein  $1/4=1/|\mu_4|$ ) are mutually consistent, so *no free dimensionless Newton dial enters*. The *new* content is a coincidence: the first law needs the coefficient  $2\pi/\eta$  ( $\eta$  the entropy density), the geometry gives  $1/c_3 = |\mathbb{Z}_2|2\pi\chi(S^2)$ , and these are the *same* equation iff  $|\mu_4| = |\mathbb{Z}_2|\chi = 4$  (v216) — so the thermodynamic and geometric origins of  $c_3$  are *one*, and  $c_3$  is triply over-determined (anchor v23, geometry v58, thermodynamics v358). The matter side is assembled (the CHM ball modular Hamiltonian, boost via Bisognano–Wichmann v323, gives the flux  $\delta \langle K_B \rangle = \frac{8\pi^2 R^4}{15} \delta \langle T_{00} \rangle$ ) and the entropy density is atom-fixed ( $1/4=1/|\mu_4|$ , central charge  $c = g_{\text{car}} + N_{\text{fam}} = 8$ ). (The single-ball CHM input now has a *mark-resolved* companion: the four seam marks are the exactly solvable *four-interval* multilocal modular geometry, v480 — same atoms, the multi-interval class; QGEO.SYM.01 stays [O].) **Full covariant equation** ([verification/v359\_grav\_nonlinear\_einstein.py]). Demanding stationarity at fixed *volume* (Jacobson 2015) brings in the EINSTEIN TENSOR  $G_{ab}$  (not just  $R_{ab}$ ; trace  $g^{ab}G_{ab} = (1-d/2)R = -R$  in  $d=4$ ) — the unique divergence-free metric tensor (Lovelock), so the equilibrium yields the *full* covariant  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  with  $\nabla^a T_{ab}=0$  an output, and TFPT fixes *both* coefficients ( $8\pi=1/c_3$  from (iii);  $\Lambda$  from  $\alpha, \rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$ , v60). So B6's full covariant field equation is now parameter-free at the *local* level. **Residual:** only the equation-of-state status (Jacobson framing, not a from-action quantisation; an *external* candidate for the missing action level — Bianconi's entropic action, Phys. Rev. D 111, 066001 (2025) — is quantified in v473:  $\beta'_B = c_3/6$  pinned,  $R^2$  gap  $3(8\pi)^9$  pre-registered as kill test, typing unchanged) and the absolute scale  $v_{\text{geo}}$  remain [O] — the global ambient measure (C7=QG.AMB.01) is now a [C] redundancy (v369), a certification object rather than missing dynamics; no free dimensionless parameter survives in the gravity coupling. So the metric sector now reads:  $R+R^2$  action (i), gap-decoupled measure (ii), and a *parameter-free* full covariant classical field equation (iii); the ambient nonperturbative measure is a [C] redundancy, not an open structural hole. (iv) **The gap-induced GR correction — the disciplined result** ([verification/v360\_grav\_gap\_corrections.py]). Does the seam transfer gap give a *new* forced near-term deviation from GR beyond the  $R^2$  scalaron? Run with the v354/v355 discriminator, the honest answer is *no*: the leading higher-curvature term *is* the  $R^2$  scalaron ( $M_{\text{scal}}^2/M_{\text{Pl}}^2 = c_3^7$ ), which is the next-order term of the same entanglement- equilibrium route (the Wald-entropy first law  $\rightarrow f(R)=R+R^2$ ) and is *already* a live CMB-S4 kill test ( $n_s, r$ ). A *distinct* gap-induced curvature<sup>2</sup> term enters only at the seam/Planck scale ( $\Delta = 6 \ln \frac{3}{2}$  is a dimensionless rate, so the scale is  $\xi v_{\text{geo}}$ ), and a new dimensionless coefficient there is *not* atom-forced (the scalaron exponent  $7=g+N-1$  and the gap exponent  $6=2N_{\text{fam}}$  are different structures) — so it is [O] declined, not manufactured. The calculable GR deviation is the scalaron. (v) **Matter–gravity backreaction** ([verification/v361\_grav\_backreaction.py]). Feeding the carrier stress tensor into the parameter-free equation: the carrier's gravitational anomaly is forced,  $c_- = 16/2 = 8 = g_{\text{car}} + N_{\text{fam}}$  (the chiral central charge of the 16-Majorana content, the same  $c_-$  as (iii)); and the matter vacuum-energy backreaction is *finite* — gap-suppressed (the Decoupling Theorem) to the forced  $\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$  (v60), *not* the  $M_{\text{Pl}}^4$  catastrophe — so the loop (the equation  $+ \Lambda +$  the carrier vacuum) closes. Beyond  $c_-=8$  and  $\Lambda$  there is [O] no new atom-forced gravitational read-out of individual SM quantum numbers (the backreaction is GR with the SM content); the discipline declines manufacturing one.

*Update (the QG.AMB.01 roadmap, consolidated — [verification/v275\_qgamb\_roadmap.py]).* The gate now carries one module that states its obligations precisely: **Tier A** (gap decoupling,  $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - 2.248 c_3^2 = 1.648 > 0$ ) makes every low-energy readout independent of the ambient measure (not a blocker); **Tier B** reduces the ambient bulk measure to identifying the seam-Calderón boundary with the holomorphic  $(E_8)_1$  net ( $c=248/31=8$ , v77), now *closed modulo cited theorems* on the MMST route SEAM.EQUIV.MMST.01 (lattice v367/v368 + the  $S_3$  stack v376–v379, Lean FORM.SEAM.MMST.01; the parent SEAM.EQUIV.01 stays [O]); the *perturbative* 4d layer is now in hand (v269/v271/v273). [C] what remains is then a *redundancy*, not missing dynamics — the nonperturbative interacting ambient measure is the Euclidean-QG conformal-factor problem (Gibbons–Hawking–Perry 1978), *inherited not TFPT-specific*; by Tier A it is gap-decoupled, so NO TFPT readout depends on it (v335/v369). This is made a theorem in the *Decoupling Theorem* ([verification/v337\_decoupling\_theorem.py]): every dimensionless readout

factors through the gapped admissible transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$ , whose finite susceptibility  $\chi = 729/665$  gap-suppresses the ambient contribution (margin  $6 \ln \frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$ ), and the negative control (gap closed  $\Rightarrow \chi$  diverges) makes it non-vacuous — so TFPT provably does not *need* the ambient measure. It is therefore reclassified from “TFPT structural frontier” to a **[C]** *redundancy* (a certification object, conditional on SEAM.EQUIV.01+Bisognano–Wichmann, v369): its obstruction is characterised (v332) and conditionally resolved by the GHP contour rotation + IDG dressing (v334); not a TFPT structural hole (QGAMB.ROADMAP.01). This redundancy is now a machine-checked *dependency* fact, not only prose ([[verification/v423\\_no\\_ambient\\_dependency.py](#)], QGAMB.NODEP.01): QG.AMB.01 is a certification *sink* — every claim that lists it is a gravity-gate/QG row, never a frozen prediction — and it is *not a value source* — it has no producing script, while  $\alpha^{-1}$ , the mixing angles and the cosmological readouts are each *computed* by a compiler script (v3/v9/v268/v7), none of them an ambient-measure script. The single transitive link is architectural, not numerical: of the frozen readouts only  $\theta_{13}$  references QG.AMB.01, through the 4d-QFT contract spine (PS.DIRAC $\rightarrow$ CONTRACT.QFT4D $\rightarrow$ QG.G2 $\rightarrow$ GATE.METRIC) — a coherence edge from living inside the unified spectral-action construction, not an input to its value  $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6}$  (v268). Honest scope: this kills QG.AMB.01 specifically; it does *not* discharge SEAM.EQUIV.01, on which the readouts still depend.

*Tier B made concrete* ([[verification/v277\\_seam\\_calderon\\_e8\\_match.py](#)]). The seam-Calderón  $\rightarrow (E_8)_1$  identification is reduced to a *single* bit. **[E]** the full  $(E_8)_1$  matching target is computed and matches:  $c=8$  (Sugawara 248/31),  $\det \text{Cartan}(E_8)=1$ , the holomorphic character  $E_4/\eta^8 = j^{1/3} = q^{-1/3}(1+248q+\dots)$  with a single primary and exactly  $248 = \dim E_8$  weight-1 currents. **[E]** the discriminator is *holomorphy*: the  $c=8$  counterexample  $(D_8)=SO(16)_1$  has  $\det \text{Cartan}=4$  (four primaries, non-holomorphic) — same  $c$ , different det, so the one discriminating condition is  $|\det K|=1$ . **[C]** the seam side matches via the flat  $\tau=i$  pillowcase (v276) and the Kummer/K3 (v260). **[C]** the residual is exactly that single holomorphy bit — now *closed modulo cited theorems* on the MMST route SEAM.EQUIV.MMST.01 (lattice v367/v368 + the  $S_3$  stack v376–v379, Lean FORM.SEAM.MMST.01; cited continuum existence **[O]**, v336) (QGAMB.TIERB.01).

*The modular-data view* ([[verification/v281\\_holomorphy\\_modular\\_data.py](#)]). The Tier-B bit is the anyon count:  $\#\text{anyons} = |\det \text{Gram}|$  ( $E_8 \rightarrow 1$  holomorphic,  $SO(16) \rightarrow 4$ ), the condensation tower  $D_5 \oplus A_3(16) \rightarrow D_8(4) \rightarrow E_8(1)$  each step a Lagrangian  $Z_2$  boson, and Gauss–Milgram returns  $c=8$  — so the open bit is simply “does the seam net condense to  $\det K=1$ ” (QGAMB.MODULAR.01).

**The live kill-test scorecard** ([[verification/v283\\_live\\_killtest\\_scorecard.py](#)]). The recent round’s computable predictions confronted with current data:  $\sin^2 \theta_{12}$  ( $+0.31\sigma$ ),  $\sin^2 \theta_{13}$  ( $+1.45\sigma$ ),  $J_{\text{PMNS}} = -0.0297$ ,  $\Sigma m_\nu$  floor 0.0586 eV, proton decay  $\tau_p = 1.55 \times 10^{35}$  yr — all consistent at  $< 2\sigma$ , with two sharp near-term kill tests ( $\Sigma m_\nu$ : DESI/CMB-S4; proton decay: Hyper-K) (PRED.LIVE.01). Two *named post-hoc* **[O]** candidates sit next to this board without touching the frozen record: the third-generation ladder-base pattern (the three  $\sim 2\sigma$  mixing tensions  $\theta_{13}/\theta_{23}/|V_{cb}|$  as one common  $-\varphi_0^{\text{ret}}$  suppression, FLAV.THIRDGEN.PATTERN.01, v467) and the splitting-ratio relation  $\Delta m_{21}^2/\Delta m_{31}^2 = |J_{\text{PMNS}}|$  at  $-0.19\sigma$  (FLAV.DM2RATIO.01, v468) — both pre-registered with symmetric kills (JUNO / Belle II), see document 2.

*Update (QBL chain, v108–v113)*: the boundary-net residual now carries *double* weight — the carrier choice  $g = 5$  has been reduced to the *same* premise. The certified scalar kernel exists iff it pairs the two sheets, the certified channel counts the code identically (Pascal closure = channel identity), pair transport is minimally complete, and the single quasi-free kernel (rank =  $c$ : 5 carrier block, 8 seam hull) is the defining datum of the free  $c = 8$  net this gate already posits. Closing the index-4 statement therefore also closes the carrier selection **[E]** (see `tfpt_1` and `tfpt_research_contracts`). [[verification/v113\\_quasifree\\_kernel.py](#)]

*Update (free-bulk premise discharged, v160–v165)*: that “free  $c=8$  net” premise is no longer free-standing. It is a fixed-point theorem (quasi-free  $\Rightarrow$  all higher cumulants  $\kappa_{2n}=0$ , v160); the infinite Schwinger cone is *eliminated* — by Bogoliubov second quantisation the cone gap equals the one-particle gap  $(2/3)^6$  (v161), forced by  $\mu_4$ -equivariance (v162); and the residual factors into the already-open  $A_2$  net-existence (an assembly of established net constructions, **[C]**) and  $R_1$  is the seam-realisation keystone SEAM.EQUIV.01, with the irreducible core  $\{\pi, v_{\text{geo}}\}$  a theorem (v165). So the metric-sector gate carries *zero new* independent open content. [[verification/v165\\_premise\\_a\\_closure.py](#)]

*The deduction below the premise is Lean-formalised (FORM.QGEO.02/FORM.QGEO.01)*. After the residual reduces to the single seam-realisation postulate QGEO.SYM.01 (the carrier  $\mu_4$  clock is the conformal deck of the seam; `tfpt_research_contracts`), its geometric normal form (the UNIFORM node:  $z \mapsto iz$ ,  $\sigma\rho\sigma = \rho^{-1}$ ,  $\text{orbit} \rightarrow \mu_4$ , multiplier order-4  $\Leftrightarrow \zeta = \pm i$ ) and the conditional im-

plication  $\text{mark-local} \Rightarrow \omega \circ \rho = \omega$  are machine-proved in the carrier-rigidity Lean project (AUDIT: PASS, only the three standard kernel axioms; mirrors `[verification/v177_seam_marking_kernel.py]` and `[verification/v210_mark_local_dtn.py]`). So everything *downstream* of the postulate is [E]; QGEO.SYM.01 *itself* is the downstream face of the keystone SEAM.EQUIV.01, which is now *closed modulo cited theorems on its MMST route* SEAM.EQUIV.MMST.01 (lattice +  $S_3$  stack, Lean FORM.SEAM.MMST.01; only the cited continuum existence stays [O] — its extension leg since re-founded on peer-reviewed crossed-product theorems, the realisation input reduced to invariant level  $R1'$ , v469; the twistor route SEAM.EQUIV.TWISTOR.01 and the unconditional parent stay [O]).

A second, independent route (`[verification/v207_asymptotic_safety.py]`, [O]). Beside the seam-net programme, the boundary data also support a **continuum** attack: an FRG truncation with an independent torsion field (the old UFE truncation) has a non-Gaussian UV fixed point —  $\partial_t g = (2 + \eta_R)g$  is UV-repulsive at  $g \rightarrow 0$  and damped at large  $g$  by the positive-definite torsion  $K^2$  term, so  $2 + \eta_R = 0$  has a root  $g_* > 0$  (intermediate value), with the scale-invariant axion coupling fixed and small,  $y^2 = 16c_3^2 = 1/(4\pi^2) = 0.0253$ . This is recorded as a *plausibility cross-check* that the QG gap is reachable from a second direction — it is **not** load-bearing and does *not* close  $G_{\text{metric}}$ ; the primary, machine-checked route stays the seam-net. A third, quantitative approach to the keystone — the celestial and twistor continuum route, now named SEAM.EQUIV.TWISTOR.01 (contract CELEST.SEAM.01, WP1–WP5d, both WP5d stages, plus the WP5e  $\alpha, \beta, \gamma, \delta_1, \delta_2, \varepsilon_2$  and  $\varepsilon_1$  stages and the M1–M3 back-reaction milestones executed: the level-2 null ideal derived, its deleting operator explicit, the GNS limit state constructed, the two-interval index chain measured, split + strong additivity witnessed on the lattice (all three KLM ingredients of complete rationality; the continuum uplift is Xu’s theorem, cited not claimed), the  $q^{-1/3}$  prefactor + level  $k=1$  pinned on the CFT side, the equivariant anomaly ledger exact on the twistor side (index bridge,  $-78$  defect, rigid  $32T_3$  residual, the honest  $a_0$  refutation), the sphere-axion exchange sub-branch closed with a certificate, the collapse confirmed full-tensorially with the one cubic  $d$ -symbol door open (its  $c_d = 1920 = |W(D_5)|$  fingerprint honestly typed look-elsewhere-loaded and convention-contingent, the stable core  $c_d = 32 \times 60$ , v513), “one level” lifted to a theorem of clock invariance with the sector counter pinning  $k = 1$  (v508/v509/v511/v513), the  $O(-2)$  bulk-axion slot built with  $\tilde{\lambda} = 6$  triply pinned (v514), the back-reacted  $\Omega_N$  closed-form with  $(2\pi i)^2$ -integral lockstep periods and the lens-forced source charge  $4 = |\mu_4|$  (v515), the twisted KS measure on the declared completion reading cancelling the  $32T_3$  residual and supplying the  $\psi = 64$  slice without the cubic  $d$ -channel (v516), the  $a_0$  uplift coupled to the centre count on four scales, monodromy  $i$  (v517), the  $\delta_1$  chain decided — kill under the derived measure, the declared/derived measure tension stated honestly (v518) — the measure question since decided at probe level for the declared reading (single-valuedness derived from F-independence + the Quillen pairing under the typed premises TP-1..TP-4, v520), and the  $w_m$  normalisation since derived constructively at probe level ( $1/\det_j$  = the Atiyah–Bott/zeta-determinant fixed-point factor, three independent sources, v523); WP5e proper [O], the global BCOV quantisation — narrowed to the global BCOV integral beyond the fibre zero-mode factor — the single remaining milestone) — is synthesised as a dedicated section in `tfpt_3_e8_audit_bootstrap`; it does not close SEAM.EQUIV.01 either.

### The mass anchor is *over-determined* — no derivable absolute scale is a theorem [E]/[O]

The single dimensionful input ( $v_{\text{geo}}$  = the reduced Planck mass  $M_{\text{Pl}}$ ) is not a hidden gap but the *unit*, and it is over-determined (`[verification/v274_scale_overdetermination.py]`). [E] every dimensionful output is a pure number  $\times M_{\text{Pl}}^d$ :  $\rho_\Lambda/M_{\text{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$ ,  $M_{\text{scal}}/M_{\text{Pl}} = c_3^{7/2}$ , the whole spectrum — the absolute  $\nu$ -scale (v272) and  $v_{\text{geo}}$  are the *same* anchor. [C] **two independent routes agree**: gravity gives  $M_{\text{Pl}} = (8\pi G)^{-1/2} = 2.4353 \times 10^{18}$  GeV, and inverting the compiler’s dark-energy prediction with the measured  $\rho_\Lambda^{1/4} = 2.24$  meV gives  $M_{\text{Pl}} = 2.438 \times 10^{18}$  GeV — the same Planck mass to 0.11% (*conditional* on the  $\Lambda$ -branch readout LAMBDA.BRANCH.01, [C]; the gravity route is unconditional — red-team `[verification/redteam/rt_F_qft4d.py]`). [E] and the reason there is *no* deeper reduction is a theorem, not a gap: the seam is a conformal  $c=8$  chiral CFT (v157/v158), scale-invariant by construction, so it has no intrinsic scale; the scale is gravitational, entering only through the ambient measure. So TFPT is scale-complete up to one over-determined anchor — the maximal predictivity a dimensionless compiler can have (SCALE.OVERDET.01).

## Summary

| Item            | Status  | Honest handle / what is missing                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\eta_B$        | [C]     | downstream readout = $6.1 \times 10^{-10}$ from closed $\Omega_b h^2$ ; leptogenesis interface, cleanest scenario $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda$ , $\tilde{m}_1 = m_3/A_\Lambda$ (shared decouple, no seesaw scale)                                                                                                                                                                                                                                                           |
| $m_p/m_e$       | [C]     | cross-sector (QCD scale / closed $m_e$ ); typed runnable solver with $\pm 7.5\%$ uncertainty budget (v374), not a compiler power                                                                                                                                                                                                                                                                                                                                                                           |
| Koide $Q$       | [C]     | source ratio $Q = 0.664$ near $2/3$ ; the source→pole factor $53/54$ is an <i>exact operator readout</i> of the missing $F=R+Q$ Sheet-Diamond corner [E]; the physical pole transfer ( $F_{\text{pole}}$ ) stays [C]                                                                                                                                                                                                                                                                                       |
| dark matter     | [C]     | determinant-line axion: $f_a = M_{\text{scal}}/128$ , $m_a \approx 23.8 \mu\text{eV}$ . <i>Post-inflationary</i> ( $H_{\text{inf}}/f_a \approx 66$ , isocurvature): the misalignment angle is a horizon average ( <i>not</i> observable), the relic follows from $f_a$ (in the post-inflationary window), domain-wall-safe via $N_{\text{DW}} =  C  = 1$ (v470/v472); the $3\pi/5$ spine value is retained as a structural rational, not a relic prediction — branch-level kill test, not a theory closure |
| muon $a_\mu$    | [C]     | seam vertex $a_\mu^{\text{seam}} = \delta_2/(2\pi) = 45/(524288\pi^9) = 2.879 \times 10^{-9}$ ; exact <i>value</i> , but the vertex projection is the [C] bridge ( $0.81\sigma$ dispersive, $\sim 1.5\sigma$ lattice)                                                                                                                                                                                                                                                                                      |
| quantum gravity | [E]/[C] | local field eq. parameter-free ( $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ , v358/v359); $R + R^2$ heat-kernel grounded (G2); perturbative graviton unitarity established (Stelle ghost a truncation artefact, v304/v370/v380); ambient measure (G6) a [C] redundancy (v369)                                                                                                                                                                                                                             |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in verification/status\_ledger.csv.

**Net (honest).** Two of the five have a genuine quantitative handle that already lands on the data ( $\eta_B = 6.1 \times 10^{-10}$  via the closed  $\Omega_b$ ; the axion DM candidate with a closed misalignment angle). One is a computed *near-miss* reported without tuning (Koide 0.664 vs  $2/3$ ). One is a typed runnable transfer ( $m_p/m_e$  as a cross-sector ratio, v374); quantum gravity has been sharpened — its local field equation is now *parameter-free* (the full covariant  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  from fixed-volume entanglement equilibrium, v358/v359), its  $R + R^2$  form heat-kernel grounded (G2), its physical sector gap-decoupled (G5), and *perturbative* graviton unitarity established (the Stelle ghost a truncation artefact, v304/v370/v380; the entire-form-factor graviton-exchange amplitude  $e^{-p^2/M^2}/p^2$  is finite, UV-softened and tree-unitary, [verification/v386\_grav\_amplitude.py], and UV-finite at every loop order by power counting — the super-polynomial falloff drives the superficial degree of divergence to  $-\infty$ , regulated by the seam scale  $M_{\text{scal}}$ , [verification/v389\_grav\_loop\_finiteness.py], with full loop *unitarity* of the nonlocal form factor honestly the still-open point; the nonlocal action's IR projection *is* the atom-scaled  $R + R^2$ ,  $M_{\text{scal}}^2/M_{\text{Pl}}^2 = c_3^7$ , so the local readout is its shadow, not a truncation with sawn-off legs, [verification/v400\_grav\_nonlocal\_action.py]), leaving the ambient projective *measure* (G6) as a [C] redundancy (v369), not an open hole. None has been forced onto the compiler ladder — which is exactly the right call: the strong results stay strong precisely because the weak ones are not oversold.

**The  $F_{\text{transfer}}$  functor as typed, runnable solvers.** The four frontier transfers are one functor — compiler *source* → measured observable via cited standard physics — now promoted from contracts to runnable solvers with kill tests, each [C] (never a compiler power). *Koide* ([verification/v371\_ftransfer\_pole.py]): a clean *negative* — standard QED running cannot be the source→pole transfer (it pushes  $Q$  *above*  $2/3$ ; the required  $\epsilon$  is negative while QED's is positive), so the transfer is the non-QED  $53/54$  operator /  $(2/3)^6$  Moebius generator (v183/v82).  $\eta_B$  ([verification/v372\_ftransfer\_boltzmann.py]): the integrated BDP Boltzmann ODE at the frozen  $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda$  gives  $\eta_B$  within a factor  $\sim 1.1$  of  $6.1 \times 10^{-10}$  with no  $M_R$  dial. *Axion* ([verification/v373\_ftransfer\_relic.py]): the finite- $T$  misalignment ODE *decides* the branch — the frozen spine angle  $3\pi/5$  lands  $\Omega_a h^2$  on  $\Omega_{\text{DM}}$  (in band) while the hilltop  $170^\circ$  over-produces ( $\sim 5.4 \times$ );

in the natural post-inflationary scenario ( $H_{\text{inf}}/f_a \approx 66$ ) the angle is instead a horizon average and the relic follows from  $f_a$  with  $N_{\text{DW}}=1$ .  $m_p/m_e$  ([`verification/v374_fttransfer_qcd.py`]): the observed 1836.15 sits inside the propagated  $\pm 7.5\%$  band of the two declared externals ( $\alpha_s(M_Z)$ ,  $C_p$ ), with lattice tightening as the kill. Each remains **[C]** — a transfer target, not a forced compiler number.

## Appendix: computational verification

These are the *frontier* items: most carry **[O]** (open) or **[C]** (conditional) and are therefore *not* part of the pass/fail suite. The one quantitative handle that is reproduced is  $\Omega_b = (1 - \frac{1}{4\pi})\varphi_0^{\text{ret}}$  (whence the  $\eta_B$  order) in `verification/v7_gravity_cosmo.py`. The  $R + R^2$  structure and the gap-decoupling of the physical QG sector *are* machine-checked (`v28_gravity_fR.py`, `v36_spectral_action_g2.py`); the proton/electron ratio is now a typed runnable **[C]** transfer (`v374`) and the ambient projective limit (G6) a **[C]** redundancy (`v369`) — both conditional, not bare **[O]**. The compiler/SM identities they build on are checked by the suite in `verification/` (run `python verification/run_all.py`; see `verification/README.md`).

## External works built upon

The TFPT discrete core (the two axioms, the  $E_8$  glue, the integer skeleton) is self-contained and machine-checked. Its *bridges* to continuum/AQFT physics, to gravity and to the lattice/singularity picture build on established mathematics, listed here by topic; the body refers to these by author name.

### Modular theory and algebraic quantum field theory.

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- J. J. Bisognano, E. H. Wichmann, *On the duality condition for a Hermitian scalar field / for quantum fields*, J. Math. Phys. **16** (1975) 985–1007; **17** (1976) 303–321.
- K. Osterwalder, R. Schrader, *Axioms for Euclidean Green's functions* I, II, Comm. Math. Phys. **31** (1973) 83–112; **42** (1975) 281–305.
- S. Doplicher, R. Haag, J. E. Roberts, *Local observables and particle statistics* I, II, Comm. Math. Phys. **23** (1971) 199–230; **35** (1974) 49–85.
- R. Haag, *Quantum field theories with composite particles and asymptotic conditions*, Phys. Rev. **112** (1958) 669–673; D. Ruelle, *On the asymptotic condition in quantum field theory*, Helv. Phys. Acta **35** (1962) 147–163.
- R. Longo, *Index of subfactors and statistics of quantum fields* I, II, Comm. Math. Phys. **126** (1989) 217–247; **130** (1990) 285–309.
- Y. Kawahigashi, R. Longo, M. Müger, *Multi-interval subfactors and modularity of representations in conformal field theory*, Comm. Math. Phys. **219** (2001) 631–669.

### Conformal nets, vertex operator algebras and lattice constructions.

- I. B. Frenkel, V. G. Kac, *Basic representations of affine Lie algebras and dual resonance models*, Invent. Math. **62** (1980) 23–66; G. Segal, *Unitary representations of some infinite-dimensional groups*, Comm. Math. Phys. **80** (1981) 301–342.
- T. J. Osborne, A. Stottmeister, *Conformal field theory from lattice fermions*, Comm. Math. Phys. **398** (2023) 219–289, [arXiv:2107.13834](https://arxiv.org/abs/2107.13834), [doi:10.1007/s00220-022-04521-8](https://doi.org/10.1007/s00220-022-04521-8). (This is the seam scaling-limit theorem the TFPT continuum leg cites — the Koo–Saleur  $\rightarrow$  Virasoro convergence,  $c = \frac{1}{2}$ , and the smeared WZW-current Theorem 5.1. The internal TFPT code-token “MMST” and the legacy claim-ids `SEAM.MMST.*` / filenames `v*_mmst_*` denote *this* Osborne–Stottmeister theorem.)
- V. Morinelli, G. Morsella, A. Stottmeister, Y. Tanimoto, *Scaling limits of lattice quantum fields by wavelets*, Comm. Math. Phys. **387** (2021) 299–360, [arXiv:2010.11121](https://arxiv.org/abs/2010.11121), [doi:10.1007/s00220-021-04152-5](https://doi.org/10.1007/s00220-021-04152-5); and *Operator-algebraic renormalization and wavelets*, Phys. Rev. Lett. **127** (2021) 230601. (The operator-algebraic (wavelet/Wilson–Kadanoff) renormalization *framework* the Osborne–Stottmeister theorem above builds on.)

- M. S. Adamo, Y. Moriwaki, Y. Tanimoto, *Osterwalder–Schrader axioms for unitary full vertex operator algebras*, [arXiv:2407.18222](#) (2024).
- M. S. Adamo, L. Giorgetti, Y. Tanimoto, *Rational and non-rational two-dimensional conformal field theories arising from lattices*, [arXiv:2506.01008](#) (2025); *Representations of conformal nets associated with infinite-dimensional groups*, [arXiv:2508.07109](#) (2025).
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### Lattices and quadratic forms.

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